DO NOW — RECALL

Lesson 118 · Extreme Programming and RAD

Name: _____ Class: _____ Date: _____

1. Which lifecycle uses relatively fixed sequential stages?

2. For the binary value 1010, what is its decimal value?

3. A linked-list node has data and what other essential field?

4. In a BST, where is a value less than the current node placed?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Waterfall.

2. Model answer

10.

3. Model answer

A link/reference to the next node (for a singly linked list).

4. Model answer

In its left subtree (under the chosen no-duplicate convention).

By the end, you can…

• Test-driven development, pair programming, refactoring and continuous integration.

• Rapid application development uses prototyping and stakeholder collaboration.

• Consider user availability, architecture, maintainability and regulatory oversight.

Vocabulary — know these terms

XP — Agile method emphasising disciplined engineering practices.

TDD — Write a failing test then implement/refactor.

RAD — Development centred on rapid prototyping and user feedback.

refactoring — Improving structure without changing observable behaviour.

NEW LEARNING

Read, annotate and explain

XP practices

Test-driven development, pair programming, refactoring and continuous integration.

RAD approach

Rapid application development uses prototyping and stakeholder collaboration.

Balance cost and fit

Consider user availability, architecture, maintainability and regulatory oversight.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A small team repeatedly breaks existing features after changes. Which XP practices help?

Model answer

Automated tests written before/with code (TDD) define intended behaviour, continuous integration runs regression tests on each change, and refactoring improves design without changing observable behaviour.

Check: link each claim to the scenario and use precise OCR terminology.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain test-driven development and pair programming within XP. [6]

Question 2

Describe two reasons RAD may suit a system with unclear interface requirements. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate XP and RAD for building a safety-critical medication scheduler under time pressure. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Refactoring is intentionally changing the program's externally visible requirements.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.